

# MIND OVER MATTER

## Retrieval Practices

Issue 4, December 2025

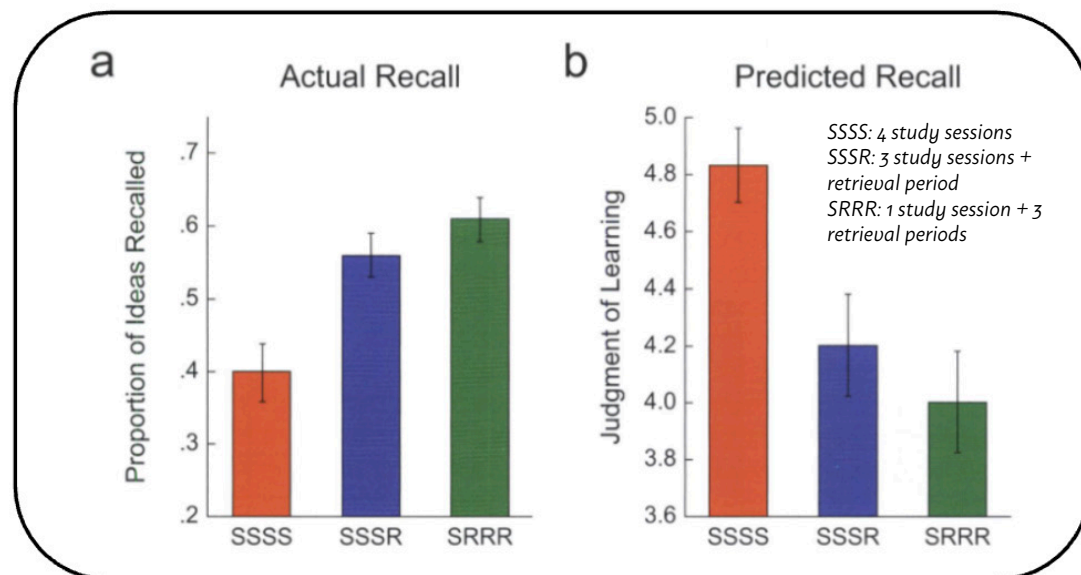
Retrieval is the **solution** to the Ebbinghaus forgetting curve. By actively retrieving information, students strengthen their memory, ensuring retention occurs long after a singular assessment is over.

Retrieval practice works because it shifts the cognitive effort from passive review to active memory reconstruction. It not only leads to memorization, it also **increases understanding**.

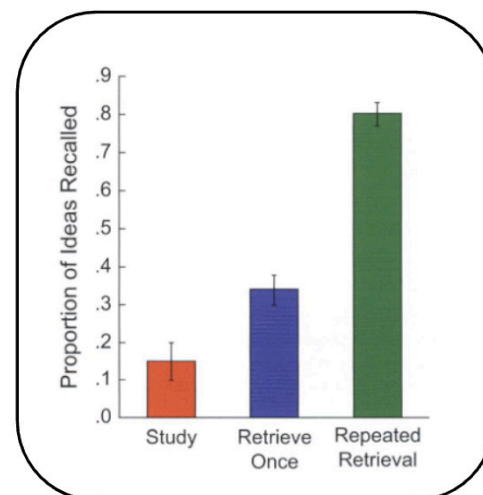
*Retrieval is a learning event, not just an assessment: it is an intervention that improves memory. The act of searching for the information consolidates the memory, making that memory/information easier to access later.*

*Retrieval improves transfer: when students retrieve knowledge in varied ways and contexts, they become better at applying that knowledge to new problems and novel situations*

*Retrieval diagnoses gaps: it provides the student and teacher with immediate and low-stakes feedback about what the student genuinely knows and what needs more work. This helps the student develop metacognition, the awareness of their own learning processes.*



Despite students predicting that re-reading leads to better retention, data, like shown here, supports the claim that retrieval is a more effective recall aid. Interestingly, students do not necessarily believe in retrieval, so we must help teach them the skill.



With consistent use of retrieval practices, we are (a) improving our students complex thinking and application skills, (b) improving our students organization of knowledge, and (c) improving our students ability to transfer knowledge to new concepts and scenarios. The following examples are some that have already been covered, but can be immediately integrated & make differences.

### Two Part Check-ins



After a short lesson segment, use a "Quiz & Correct" approach. Have students answer a question, then immediately allow them to check their notes to self-correct.



**Low-Stakes Reinforcement:** Provides immediate reinforcement for correct sequences or concepts and offers corrective feedback, which is key for foundational skills

### Question Generation



Instead of answering your questions, ask students to write down three questions that the material in front of them should be able to answer.



**Boosts Metacognition:** This forces them to focus on what's important and engage in self-reflection about their learning process

### Concept Mapping



Introduce a low-stakes activity that forces students to revisit a key concept taught three weeks ago and integrate it with their current topic



**Strengthens Connections:** This intentionally spaced and interleaved retrieval causes students to differentiate between concepts and strengthens the mental pathways for long-term retention

### Brain Dump



Start class with a one-minute "brain dump." Ask students to write down everything they can remember from the previous lesson or key concepts from a reading



**Activates Prior Knowledge:** This simple, ungraded activity forces students to pull up existing schemas, preparing the brain to connect new information, which helps the new information "stick" better

